

COMPUTATIONAL MICROSCOPY PROJECT

A month with AI agents on a Fourier ptychography problem

What was solved and what was not

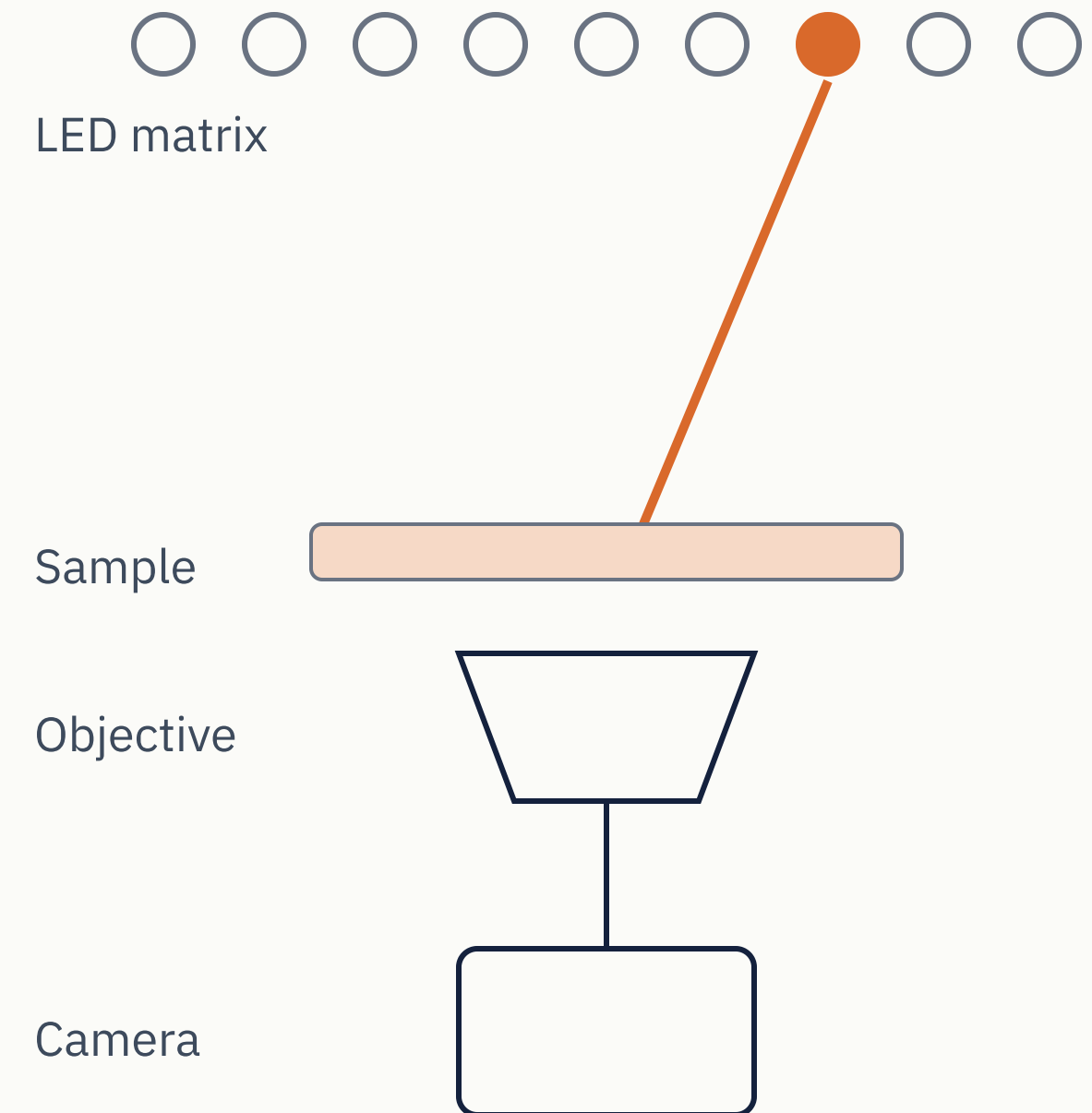
Lucas Costilla · week of 28 Sep 2026

The problem in one minute

A microscope cannot separate details below a certain limit: with our objective, **$\sim 3.8\ \mu\text{m}$** .

FPM takes $N \times N$ photos, each with the light coming from a different angle, and a program combines them to reach, in theory, **$0.5\text{--}1\ \mu\text{m}$** .

The hard part: the program has to know the exact geometry and infer something the camera does not record, how much the light is delayed (the *phase*).



The short answer

Achieved

A microscope simulator and a reconstruction program that work on simulated data

Five errors that silently gave wrong results: three in our reconstruction program and two in the validation and configuration data; one invalidated everything done with real data

The centre of the matrix relative to the axis, measured from the images (not the height: that was measured by hand)

A capture recipe: from 12 to 131 useful light angles

Recovering the phase of the real sample in all three colours, consistent with photos it did not use (28 Sep)

Not achieved

A real reconstruction with more detail than the objective that passes every test

Showing a resolution gain: there is no evidence yet, and not only because of the sample

Fully combining the three colours on real data: they show the same sample, but the fine detail does not match yet

How we worked

Direct session

One person and the assistant, step by step

Deciding, reviewing, short tasks

Parallel agents

Copies of the assistant, one task each

More than 20 audits in one week

Teams with a verifier

Lead, workers, verifier and writer

For hard questions, in rounds

One team round: the verifier redoes every number before accepting it



On simulated data, it works

With simulated data the program recovers the sample, including the phase

Made-up sample: amplitude



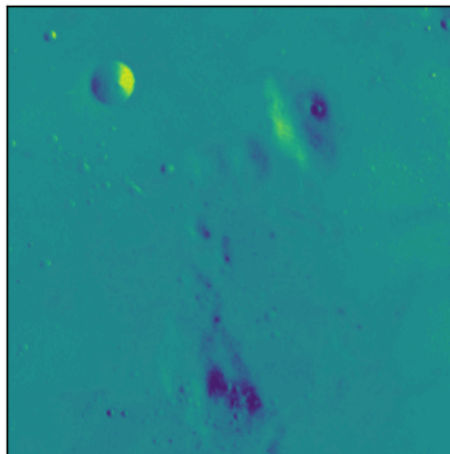
One of the 169 simulated photos



Reconstruction: amplitude

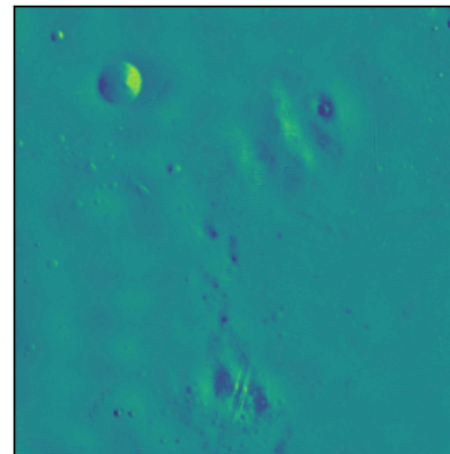


Made-up sample: phase



The phase does not appear
in any photo:
the program infers it

Reconstruction: phase



0.995

match of the amplitude (the square root of the brightness) with the original sample (1 = perfect)

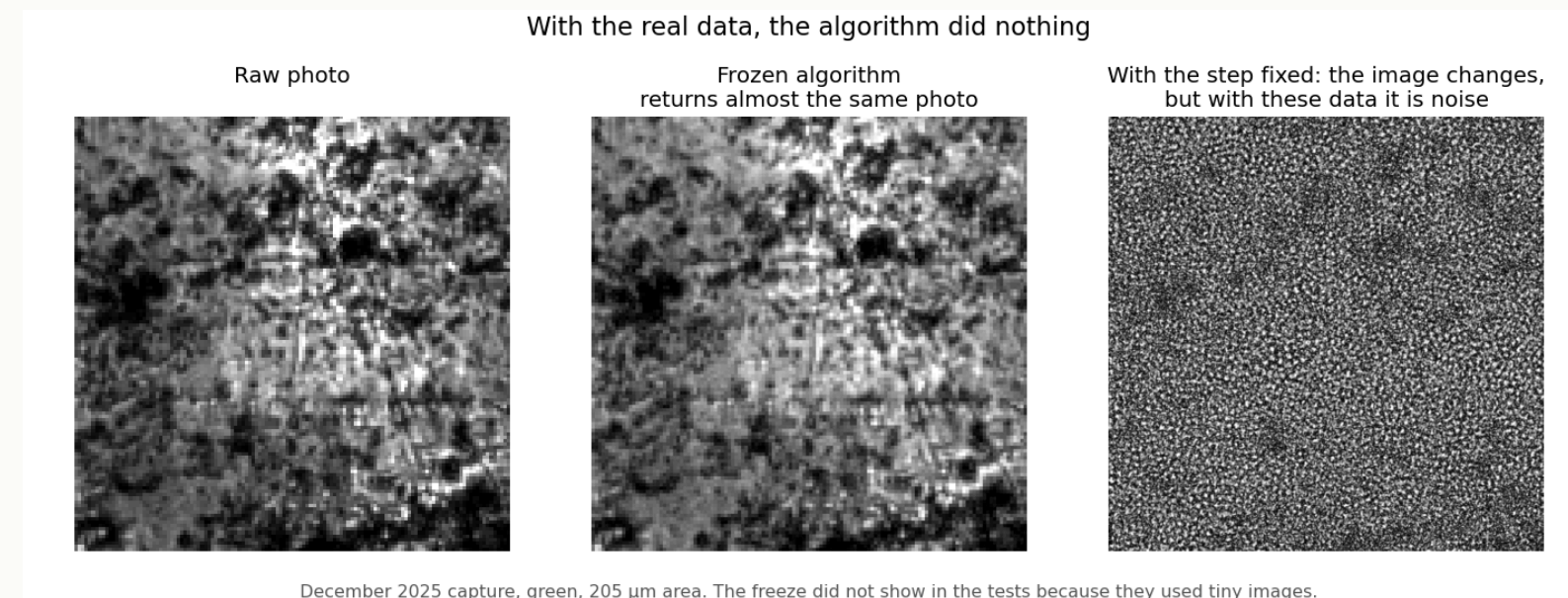
0.87

match of the phase, which no photo shows

Same geometry as our real setup, no noise (with the matrix at 100 mm, the phase rises to 0.97)

Real setup geometry (13×13 LEDs, 75 mm, 2.5× objective), no noise, 100 iterations.

The worst error: on real data, the program did nothing



The program corrects the image a little in each round. Those corrections were ~8,000 times smaller than they should have been: it returned almost the original photo.

Hundreds of automated tests passed, because they used images of 16 to 32 pixels.

An agent investigating something else found it. Everything done before on real data had to be discarded.

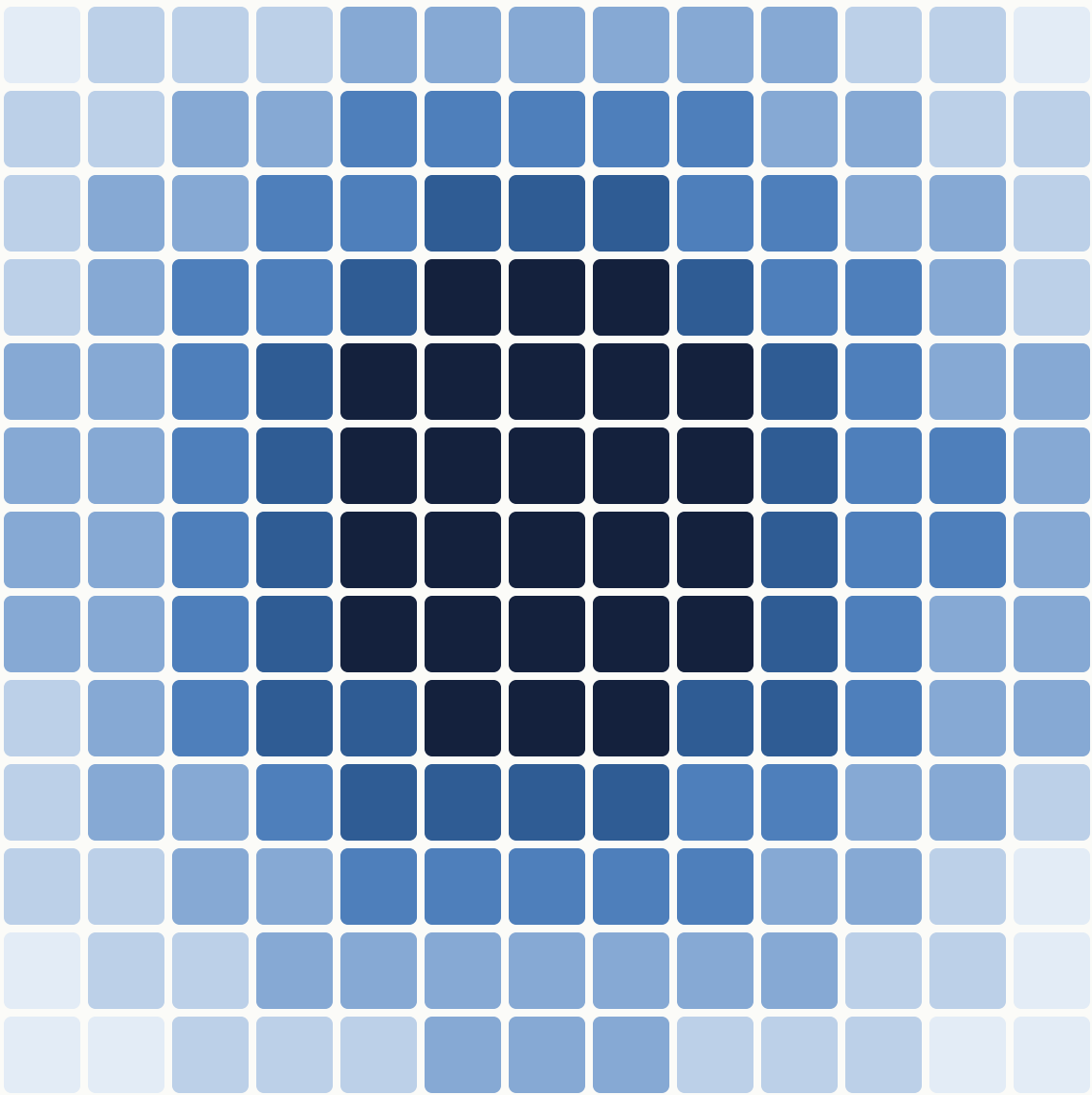
Fixing the step was necessary but not enough: with the December data the result was noise. Credible reconstructions came with the September captures.

Other errors, including the agents' own

Error	Who found it
The reference image (Zeiss, 2.5x objective) had less detail than a raw photo	An agent traced its origin
The photos were not corrected for their exposure time	An agent, while reviewing the data
The real-data runs assumed a different objective: nobody had asked	The person, when confirming the hardware
The assistant misexplained the noise in the dark frames: it was not random, it was hot pixels	The first team's verifier, by measuring

The assistant also assumed things about the hardware that turned out false: the camera, the objective, the height.
They do not know what they are not told.

The new capture: the recipe worked



Useful signal per LED, over the noise (green, 100 ms)

131 of 169

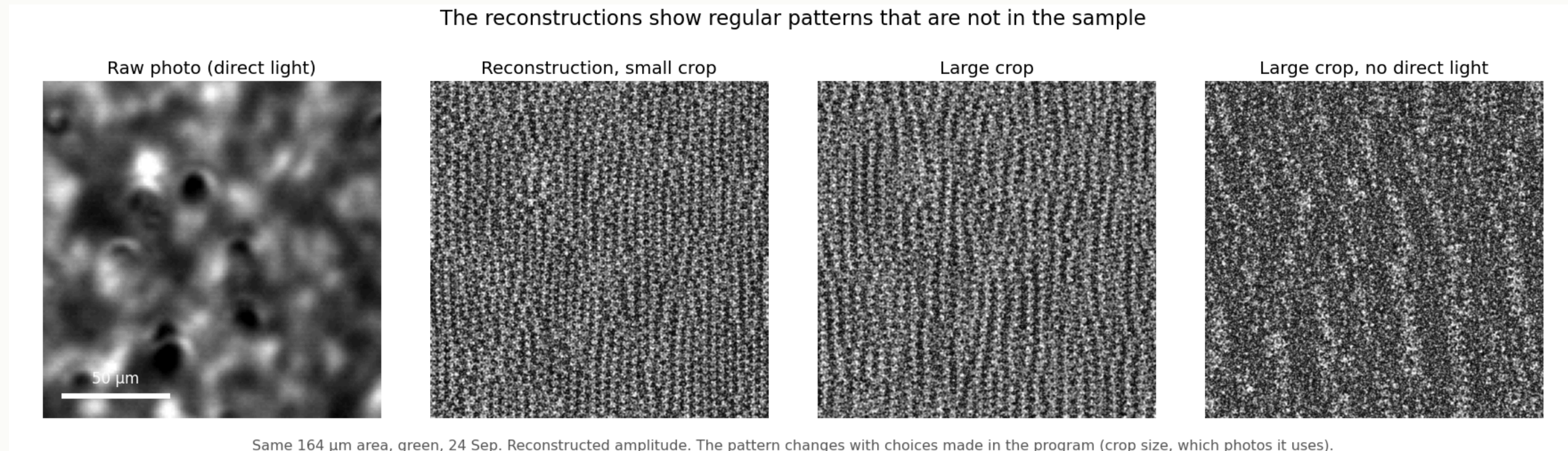
LEDs with useful signal (more than 3 times the noise)

12 of 182

LEDs with at least 10 % of the maximum brightness in July (1 ms): a different measure, but very little light

- less than 2
- 2 to 3
- 3 to 5
- more than 12

Green, 24 Sep (already with the recipe): why we do not trust that reconstruction



1. Predicts unseen photos

Passes

2. Wrong geometry shows up

Fails

3. No patterns absent from the sample

Fails by eye

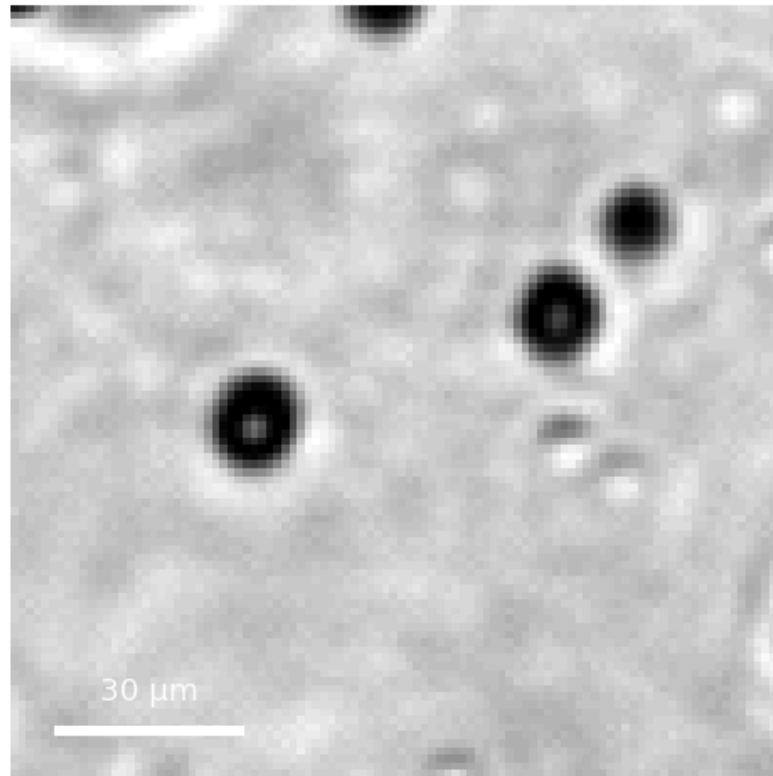
4. New detail appears

Not assessable

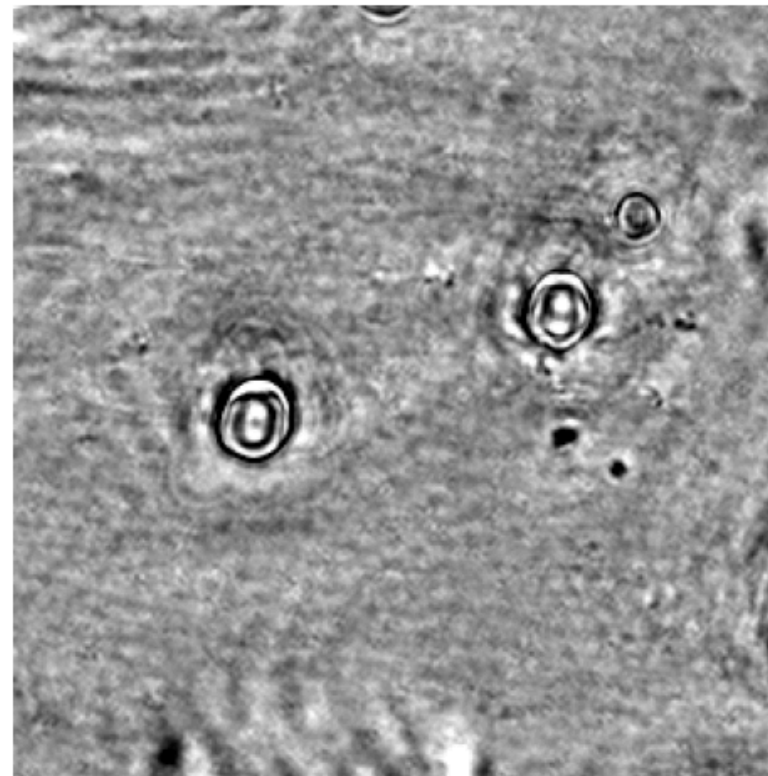
Red, 25 Sep: better image, no extra detail

Red, 25 Sep: the reconstruction shows the particles and their phase, without the grid pattern seen in green

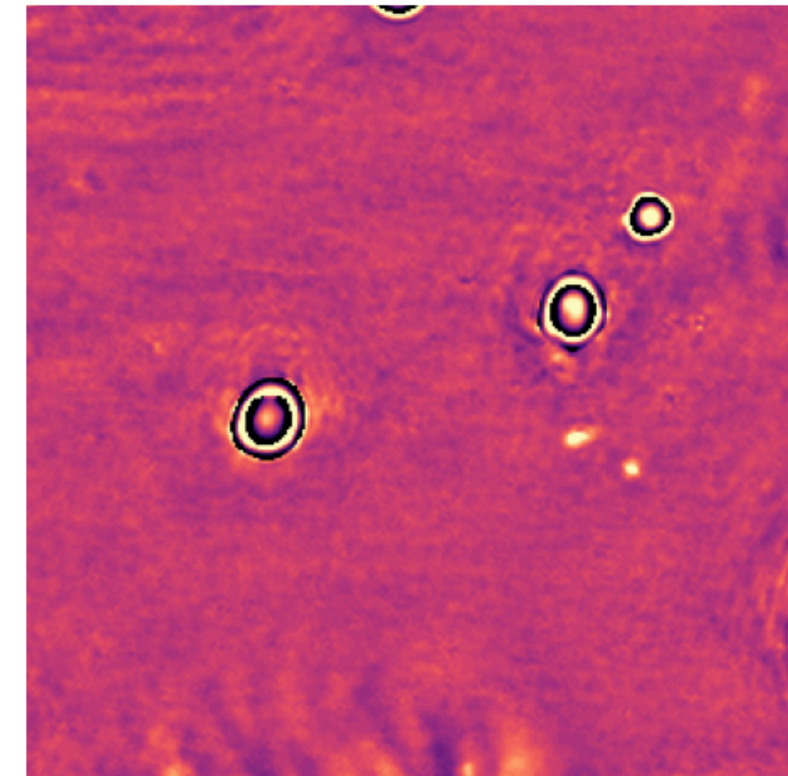
Raw photo (direct light)



Reconstruction: amplitude



Reconstruction: phase



Same 128 µm area. The phase (how much the light is delayed) does not appear in any photo: the program infers it.

1. Predicts unseen photos

Passes

2. Wrong geometry shows up

Fails

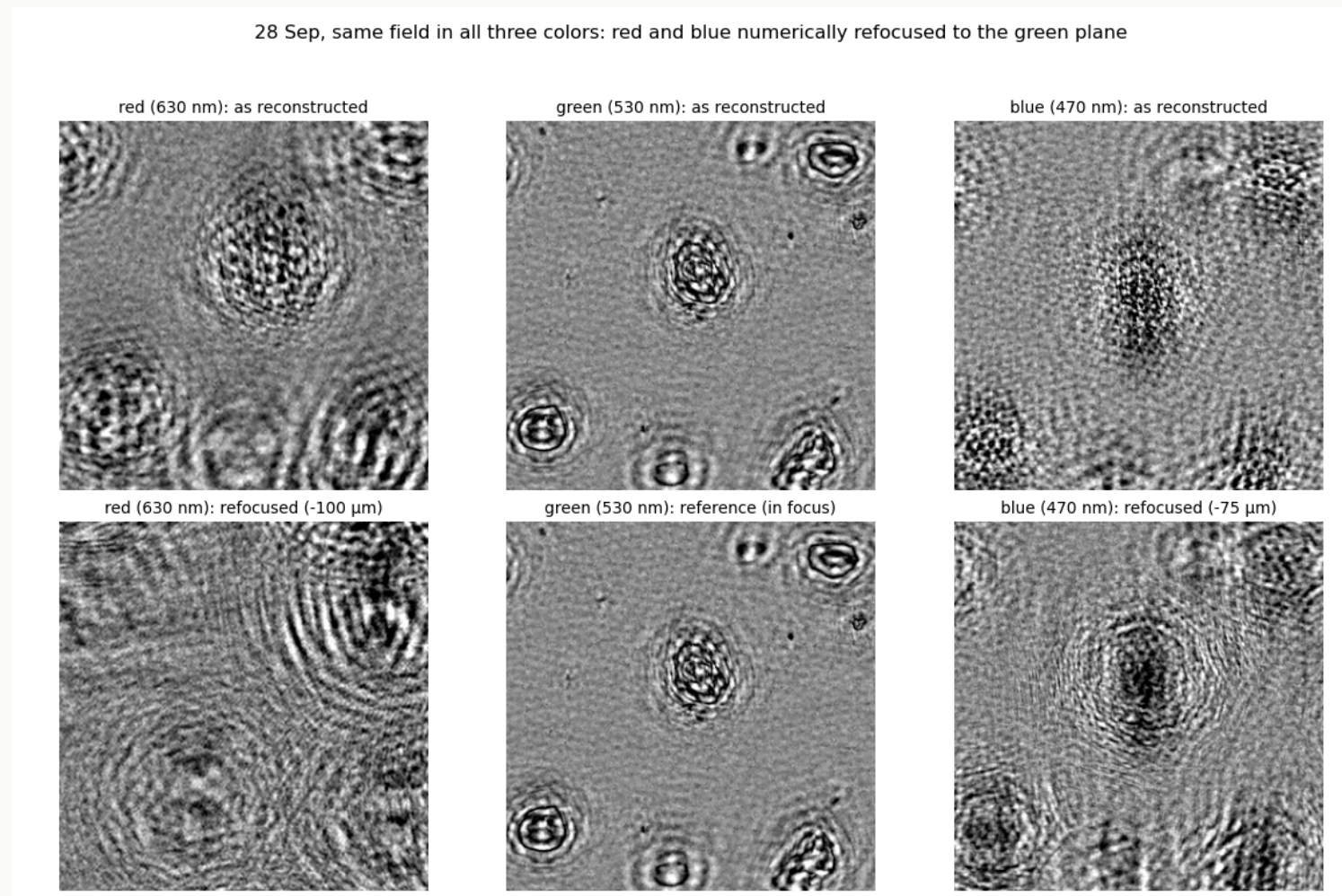
3. No patterns absent from the sample

Improves, not enough

4. New detail appears

Phase only, no new detail

Three colours, one field: no match



Raw photos across colours

Match: 0.71 to 0.78

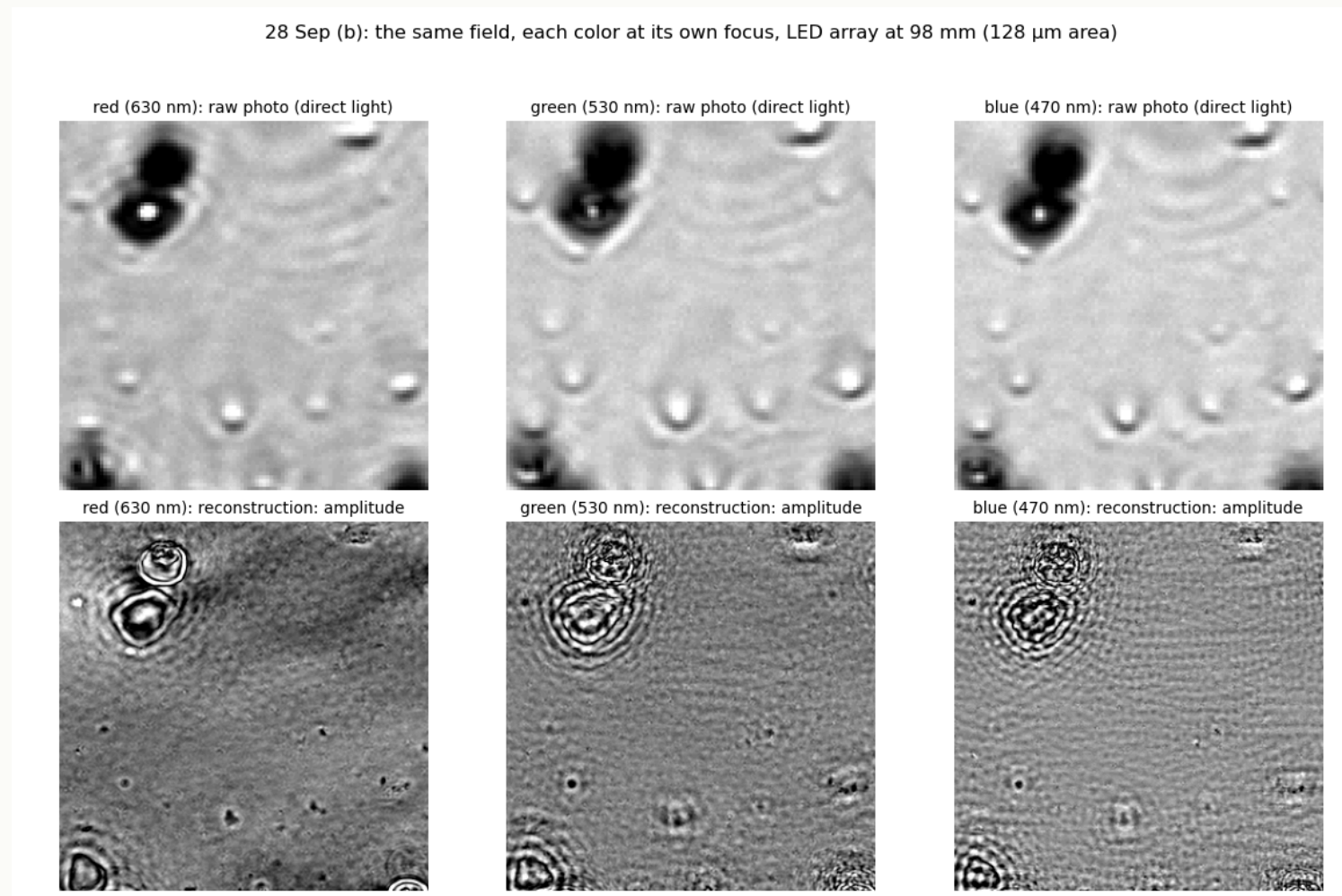
Reconstructed fine detail

No match: 0.06 at most

Only the colour in focus

looks like the sample

Each colour in its own focus: the best capture



Predicts unseen photos

All three colours: 0.46 to 0.62

Across colours

The same particles

Detail finer than the objective

Improves, not enough

Lessons for any group using agents

1

Criteria before results

Write down what counts as success before running on real data

2

Independent verification

Another agent, with different code, redoes every number

3

Every number with its source

The file that produces it, cited and checkable

4

Simulation first

Where the answer is known, under the same conditions

5

A person who knows the equipment

The agent cannot answer the hardware questions

They contribute: volume, reviewing code and numbers for errors, experiment design. **They fail at:** the physical world and overconfidence.

Next steps

1. A calibration target with lines of known width

Decides whether the method gains detail in our setup

2. Thinner samples, or a thick-sample model

The extra light comes from the sample volume (zone 1, 25 Sep)

3. Separate what improved the last capture

Overlap or per-colour focus: test each one separately

Full report: <https://claude.ai/code/artifact/de7eab31-d9e0-4828-a62d-285ba0d46a98>